POLITECNICO
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Network Data Analysis Laboratory

Proposed projects

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General requirements (1/2) – Projects tasks

- All projects **must** include the following main steps:

1. Raw data visualization/analysis
   - Does data suggest anything (ML algorithm, promising features, redundancy, …)?

2. Data preprocessing (e.g., generate «ordered» samples from raw data)

3. ML models optimization and training
   - Hyperparameters tuning with cross-validation

4. Testing the performance and evaluating different scenarios (**see specific projects assignment**)

General requirements (2/2) – Methodology

- All projects **must** include (for 12 points)
  - Specific project assignment (see later)
  - 2 or 3 ML algorithms (usually recommended in the assignment)
  - Different performance metrics: MSE, MAE, Accuracy, Precision, Recall, F-score, training duration, …

- Additionally, projects **may** include (at least) one "advanced" task (for 3 points)
  - Transfer learning: train on one domain, test on another domain (e.g., different datasets, different tasks, etc.)
  - Federated learning: compare "global" models (all data available at one location) vs. "local" models that share knowledge through Federated Learning
  - Explainability: apply XAI (eXplainable Artificial Intelligence) frameworks to interpret/explain model reasoning and validate the model
  - Project-specific advanced task

Project #1-2 – Packet Loss Event Classification
Background

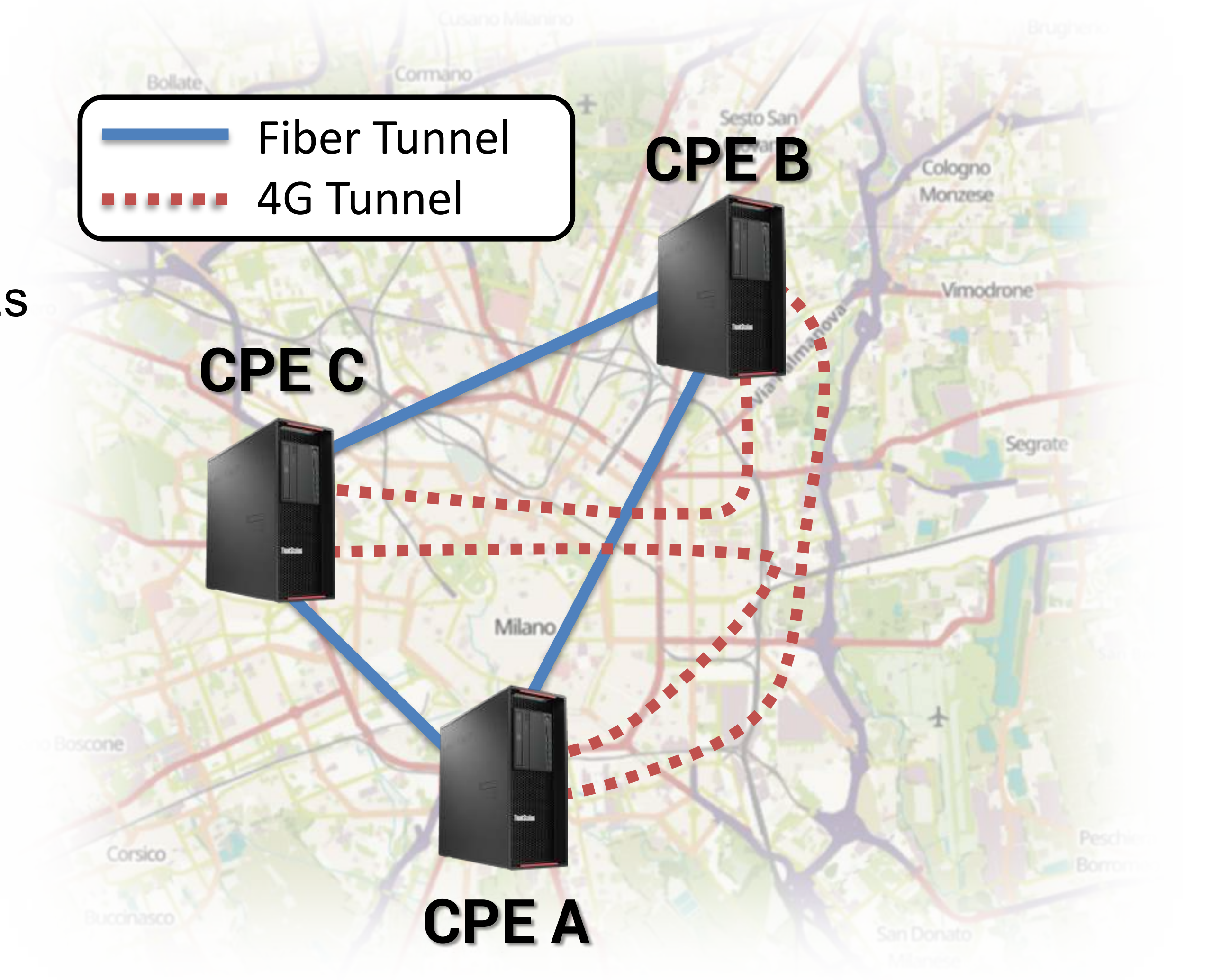
- SD-WAN (Software-Defined Wide Area Network) is an emerging technology for the interconnection of geographically distributed company branches
  - Uses **one or more public best-effort internet access** to interconnect the different sites through one CPE (Customer Premises Equipment) in each site. The interconnection is secure by conveying data through encrypted tunnels.
  - Each of these best-effort internet access corresponds to **different paths in the public Internet**

- **Problem:** Mission-critical inter-site traffic requires strict QoS (e.g., extremely low packet loss). Network inefficiencies (latency spikes, packet loss) on the paths can cause performance degradation and potential financial impact (e.g., in financial contexts).
- We would like to predict network inefficiencies (e.g. latency spikes or packet loss bursts) before they happen so as to redirect traffic on the best path

- Can we use ML to predict packet loss before it happens based on previous historical data?

Project #1-2 – Packet Loss Event Classification
Testbed setup

- Three geographically distributed SD-WAN CPEs across Milan
- Dual WAN connectivity per CPE:
  - Fiber (FTTC/FTTH) tunnel
  - 4G cellular network tunnel
- A latency measurement campaign has been performed on this setup



Project #1-2 – Packet Loss Event Classification
Dataset

- For any ordered pair (CPE X, CPE Y), two measurement files (.csv) have been generated (one for the fiber tunnel and one for the 4G tunnel)
- Two measurement windows, respectively of duration ~15h and ~ 30h, with one second granularity

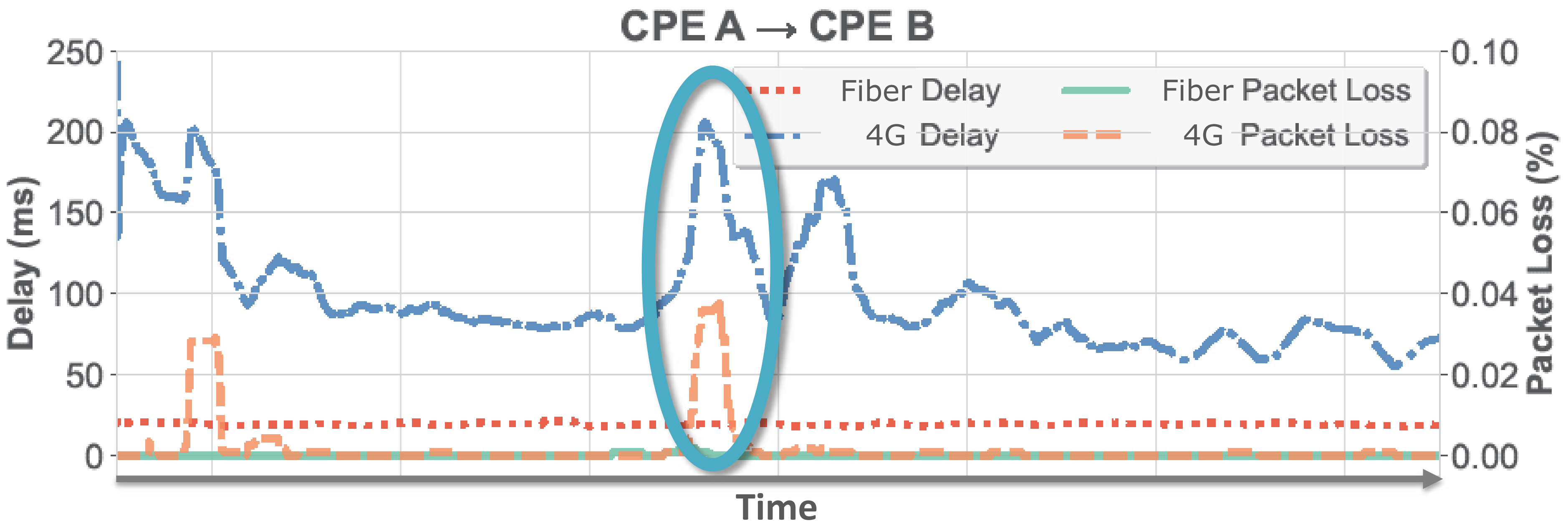
	time	delay_ms
23691	2025-03-05 19:05:22.567	13.2708
23692	2025-03-05 19:05:23.567	21.1522
23693	2025-03-05 19:05:24.566	14.8669
23694	2025-03-05 19:05:25.569	-1.0000
23695	2025-03-05 19:05:26.569	14.7499
23696	2025-03-05 19:05:27.570	13.8136
23697	2025-03-05 19:05:28.570	13.6749

Single file structure

(e.g. fiber tunnel from CPE A to CPE B)

Packet losses are identified by the value **-1** in the **delay_ms** column

Project #1-2 – Packet Loss Event Classification
The Dataset Visualized

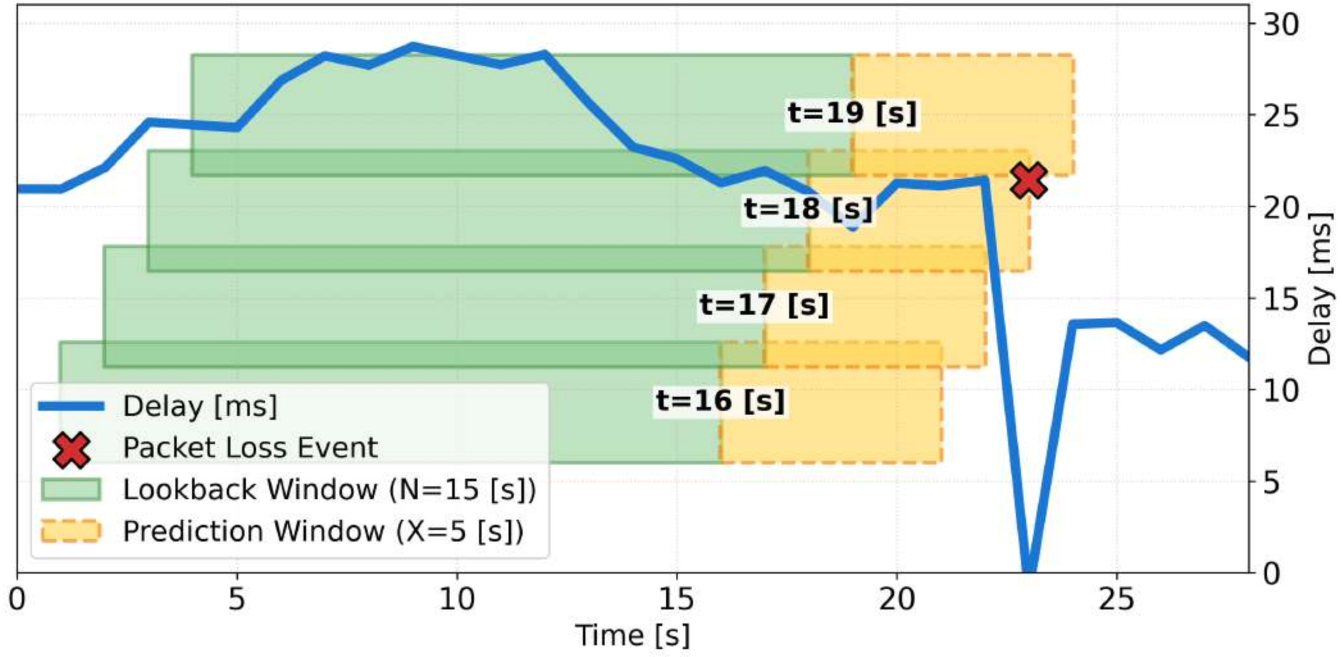


Evident **correlation** between packet loss and delay increase

Can we predict packet loss event <u>before</u> it happens looking at historical delay values?

Project #1-2 – Packet Loss Event Classification
Hints

- **Data preprocessing**
  - The rows with `delay_ms` set to `−1` (packet loss event) must be pre-processed
- **Procedure**
  - Use a sliding window approach: take the last ***N* seconds** of measurements (*Lookback Window*) as input to predict the presence of a packet loss event in the following ***X* seconds** (*Pred. Window*)
  - **Experiment with different *N* and *X* values**
  - For each Lookback Window statistical features can be used



Project #1 – Packet Loss Event Classification
Packet Loss Event Classification with RF & Transfer Learning

- (12 points) Perform classification on packet loss event using two different models
  - Neural Network
  - Random Forest
  - Evaluate feature importance
  - Experiment with different N and X values (see previous slide)
- (3 points) Advanced Task: Transfer Learning
  - Build a model trained on a single direction (e.g., CPE A → CPE B) and fine-tune the model with data on the opposite direction (e.g., CPE A ← CPE B)
  - Assess the performance of the fine-tuned model
  - Compare the accuracy between the fine-tuned model and individually computed models entirely trained on local dataset
  - Keep the last hours of each measurement windows as hold-out test set

Project #2 – Packet Loss Event Classification
Packet Loss Event Classification with XGBoost & Federated Learning

- (12 points) Perform classification on packet loss event using two different models
  - Neural Network
  - XGBoost
  - Evaluate feature importance
  - Experiment with different N and X values (see previous slide)
- (3 points) Advanced Task: Federated Learning
  - Assume a centralized entity (e.g., SD-WAN Controller) periodically collects local models (e.g., trained with data available at individual CPEs), aggregates them and redistribute the information to build a generalized model
  - Compare the performance between the general (i.e., federated) and the CPE-specific (local) models